

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

MOBILE AND WIRELESS COMMUNICATION

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Write down any two 'multiple access' techniques	M1.04	R
2	What is a 'Cell' related to cellular communication?	M1.03	R
3	Expand the term 'GPRS'	M2.04	R
4	List the main subsystems of GSM.	M2.02	R
5	The Multiple Access technology used in UMTS is _____	M3.01	R
6	State at least two differences between Wi-Fi and Wi MAX.	M3.02	R
7	Name the Telecom regulatory body in India.	M1.05	R
8	Name a popular 4G Technology	M4.01	R
9	Expand the term IoT	M4.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Explain the concept of Frequency Reuse	M1.02	U
2	Briefly explain the power control Techniques in Mobile Communication	M1.03	U
3	Differentiate between 'Soft' and 'Hard' Handoff strategy	M1.02	U
4	Discuss the authentication process in GSM Technology	M2.01	U
5	Explain what CAMEL related to mobile network is.	M2.03	U
6	Make short note on GPRS	M2.04	R

7	Discuss the features of OVSF Code	M3.01	U
8	Make note on Traffic Engineering	M1.04	U
9	Differentiate between LTE and LTE-A	M4.01	U
10	Make note on IS-95 standard	M2.04	U

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Explain the operation of Cellular System with relevant functional diagrams.	M1.03	U
OR			
2	Explain on First Generation Wireless Networks	M1.04	U
3	Briefly explain CDMA technology giving stress on the idea of orthogonal codes	M3.01	U
OR			
4	With the help of Bluetooth Protocol stack explain Bluetooth Technology	M3.04	U
5	Explain GSM Architecture	M2.01	U
OR			
6	Discuss different types of GSM logical Channels	M2.02	U
7	Explain UMTS Architecture	M3.01	U
OR			
8	Discuss in detail about mobile Wi-MAX Technology	M3.02	U
9	Explain LTE Architecture	M4.01	U
OR			
10	Make a brief note on 'Internet of Things' and its application areas.	M4.02	U
11	Explain 5G Technology	M4.04	U
OR			
12	Discuss the features and significance of LTE – Advanced Network?	M4.01	U